

COS20019 - Cloud Computing Architecture

Week 7: ACF Lab 6: Scaling and Load Balance your architecture

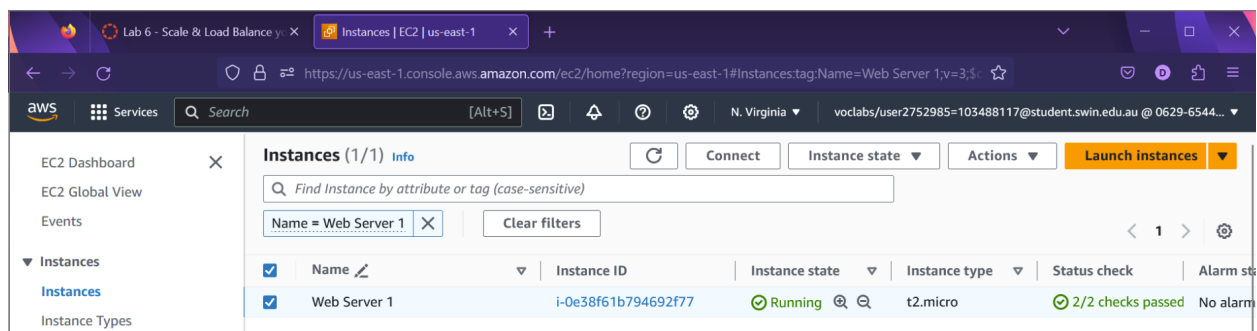
Author: Trac Duc Anh Luong - ID: 103488117

Due Date: 22/10/2023

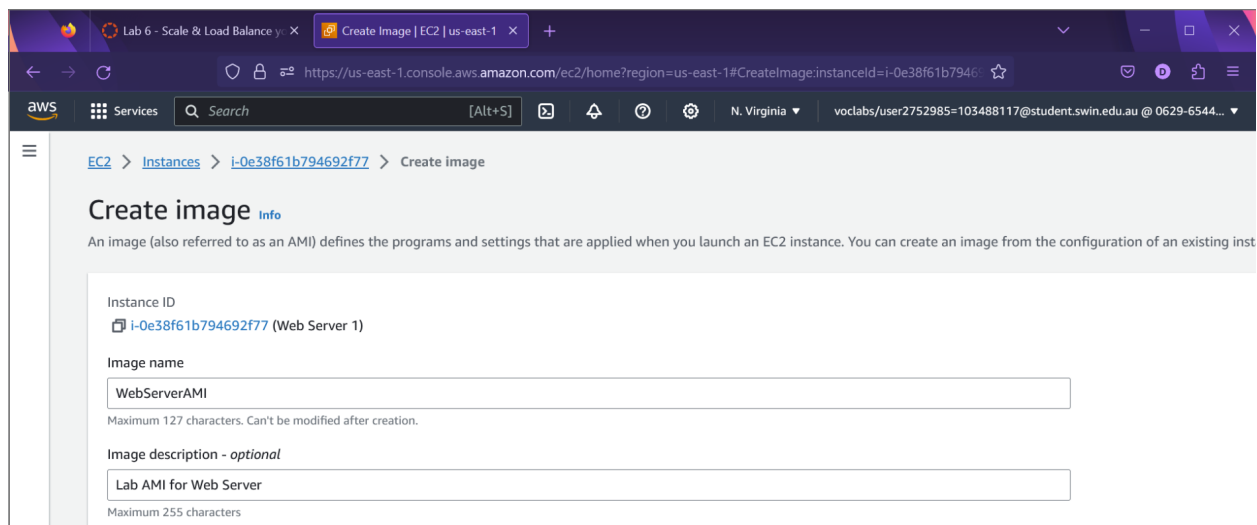
Task 1: Create an AMI for Auto Scaling

In this task, you will create an AMI from the existing *Web Server 1*. This will save the contents of the boot disk so that new instances can be launched with identical content.

5. In the **AWS Management Console**, in the search box next to **Services**, search for and select **EC2**.
6. In the left navigation pane, choose **Instances**.
First, you will confirm that the instance is running.
7. Wait until the **Status Checks** for **Web Server 1** displays *2/2 checks passed*. If necessary, choose refresh to update the status.
You will now create an AMI based upon this instance.



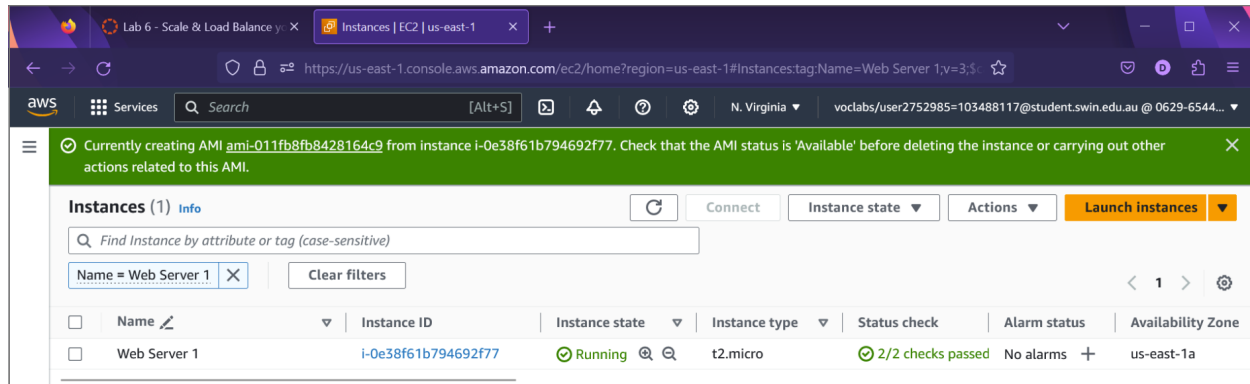
8. Select **Web Server 1**.
9. In the **Actions** menu, choose **Image and templates** > **Create image**, then configure:
 - **Image name:** WebServerAMI
 - **Image description:** Lab AMI for Web Server



10. Choose Create image

A confirmation banner displays the **AMI ID** for your new AMI.

You will use this AMI when launching the Auto Scaling group later in the lab.



Task 2: Create a Load Balancer

In this task, you will create a load balancer that can balance traffic across multiple EC2 instances and Availability Zones.

11. In the left navigation pane, choose **Target Groups**.

Analysis: *Target Groups* define where to *send* traffic that comes into the Load Balancer. The Application Load Balancer can send traffic to multiple Target Groups based upon the URL of the incoming request, such as having requests from mobile apps going to a different set of servers.

Your web application will use only one Target Group.

- Choose Create target group
- Choose a target type: **Instances**
- **Target group name**, enter: **LabGroup**
- Select **Lab VPC** from the **VPC** drop-down menu.

The screenshot shows the AWS Management Console interface for creating a target group. The browser tabs include 'Lab 6 - Scale & Load Balance', 'Step 1 Create target group | EC2', and 'Images | EC2 | us-east-1'. The URL is 'https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTargetGroup:'. The page title is 'Specify group details' with a subtitle 'Your load balancer routes requests to the targets in a target group and performs health checks on the targets.'

Step 1: Specify group details

Basic configuration
Settings in this section can't be changed after the target group is created.

Choose a target type

- ☒ **Instances**
 - Supports load balancing to instances within a specific VPC.
 - Facilitates the use of [Amazon EC2 Auto Scaling](#) to manage and scale your EC2 capacity.
- ☐ **IP addresses**
 - Supports load balancing to VPC and on-premises resources.
 - Facilitates routing to multiple IP addresses and network interfaces on the same instance.
 - Offers flexibility with microservice based architectures, simplifying inter-application communication.
 - Supports IPv6 targets, enabling end-to-end IPv6 communication, and IPv4-to-IPv6 NAT.
- ☐ **Lambda function**
 - Facilitates routing to a single Lambda function.
 - Accessible to Application Load Balancers only.
- ☐ **Application Load Balancer**
 - Offers the flexibility for a Network Load Balancer to accept and route TCP requests within a specific VPC.
 - Facilitates using static IP addresses and PrivateLink with an Application Load Balancer.

Target group name
LabGroup
A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Protocol: HTTP
Port: 80
1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.

- ☒ **IPv4**
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.
- ☐ **IPv6**
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

Lab VPC
vpc-064d3c1bcb5358a81
IPv4: 10.0.0.0/16

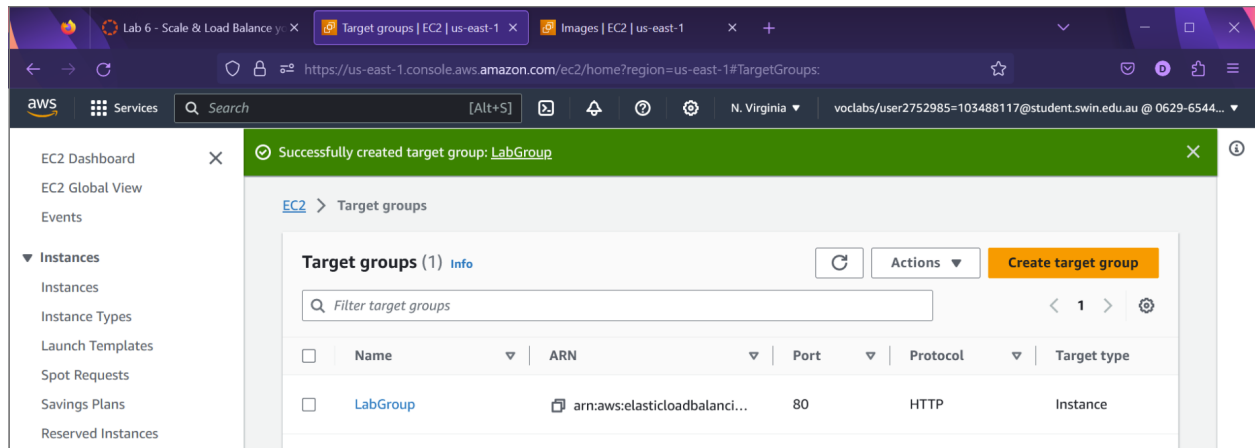
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12. Choose Next. The **Register targets** screen appears.

Note: *Targets* are the individual instances that will respond to requests from the Load Balancer. You do not have any web application instances yet, so you can skip this step.

13. Review the settings and choose Create target group

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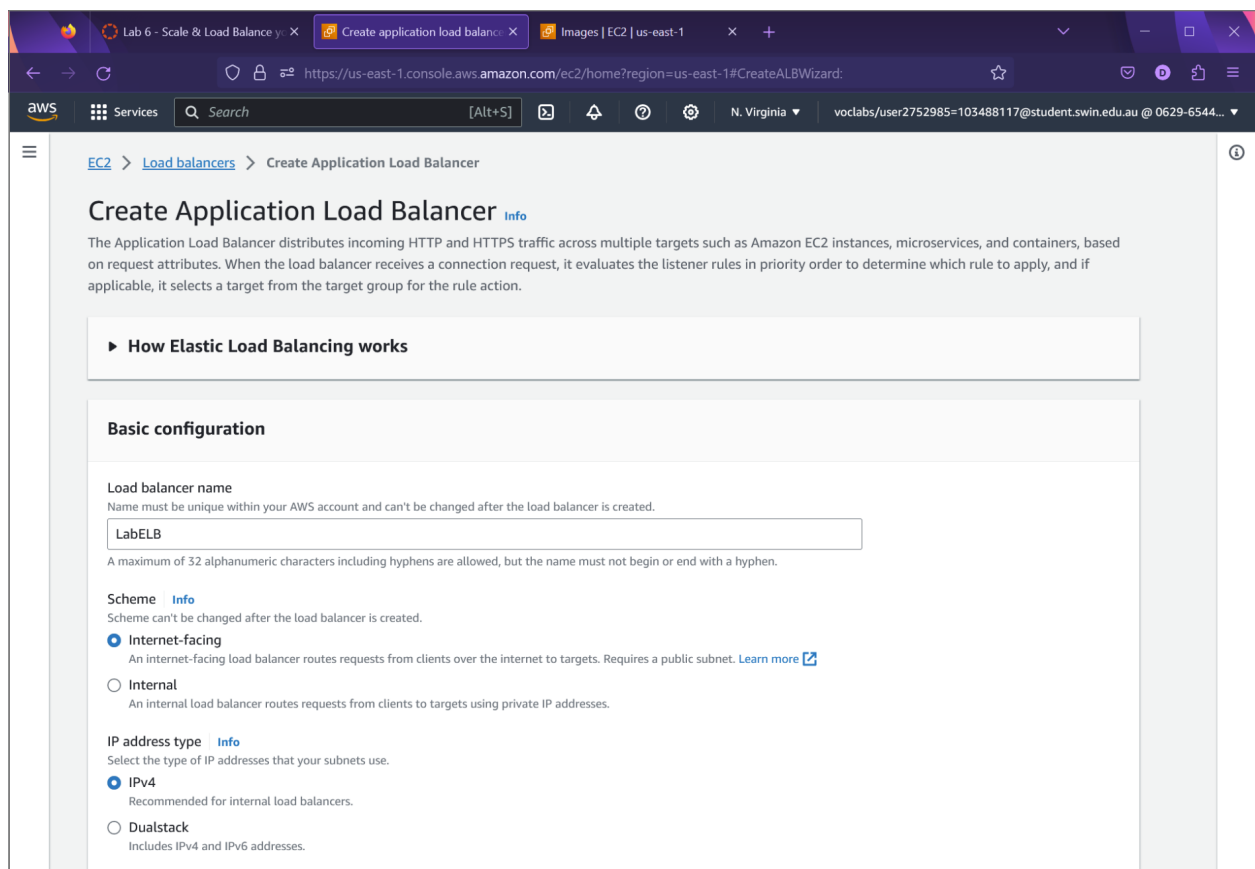
14. In the left navigation pane, choose **Load Balancers**.

15. At the top of the screen, choose Create load balancer.

Several different types of load balancer are displayed. You will be using an *Application Load Balancer* that operates at the request level (layer 7), routing traffic to targets — EC2 instances, containers, IP addresses and Lambda functions — based on the content of the request. For more information, see: [Comparison of Load Balancers](#)

16. Under **Application Load Balancer**, choose Create

17. Under **Load balancer name**, enter: **LabELB**



18. Scroll down to the **Network mapping** section, then:

- For **VPC**, choose **Lab VPC**
You will now specify which *subnets* the Load Balancer should use. The load balancer will be internet facing, so you will select both Public Subnets.
- Choose the **first** displayed Availability Zone, then select **Public Subnet 1** from the Subnet drop down menu that displays beneath it.
- Choose the **second** displayed Availability Zone, then select **Public Subnet 2** from the Subnet drop down menu that displays beneath it.
You should now have two subnets selected: **Public Subnet 1** and **Public Subnet 2**.

19. In the **Security groups** section:

- Choose the Security groups drop down menu and select **Web Security Group**
- Below the drop down menu, choose the **X** next to the default security group to remove it.
The **Web Security Group** security group should now be the only one that appears.

The screenshot shows the 'Create application load balance' wizard in the AWS Management Console, specifically the 'Network mapping' step. The browser address bar shows the URL: `https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateALBWizard:`. The page title is 'Lab 6 - Scale & Load Balance'.

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)

Select the virtual private cloud (VPC) for your targets or you can [create a new VPC](#). Only VPCs with an internet gateway are enabled for selection. The selected VPC can't be changed after the load balancer is created. To confirm the VPC for your targets, view your [target groups](#).

Lab VPC
vpc-064d3c1bcb5358a81
IPv4: 10.0.0.0/16

Mappings [Info](#)

Select at least two Availability Zones and one subnet per zone. The load balancer routes traffic to targets in these Availability Zones only. Availability Zones that are not supported by the load balancer or the VPC are not available for selection.

☒ **us-east-1a (use1-az4)**

Subnet
subnet-0287c7386b1ea16d2 Public Subnet 1

IPv4 address
Assigned by AWS

☒ **us-east-1b (use1-az6)**

Subnet
subnet-0ae5dee2c2df397d0 Public Subnet 2

IPv4 address
Assigned by AWS

Security groups [Info](#)

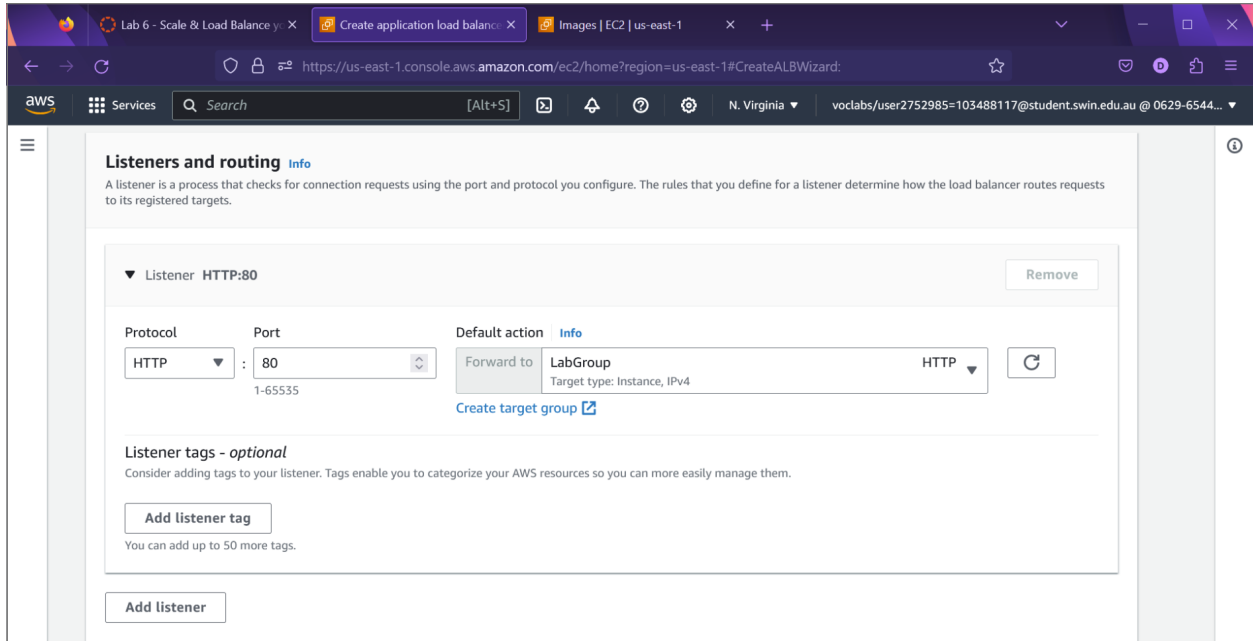
A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups
Select up to 5 security groups

Web Security Group
sg-012a22a3a6243daa4 VPC: vpc-064d3c1bcb5358a81

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20. For the Listener HTTP:80 row, set the Default action to forward to **LabGroup**.

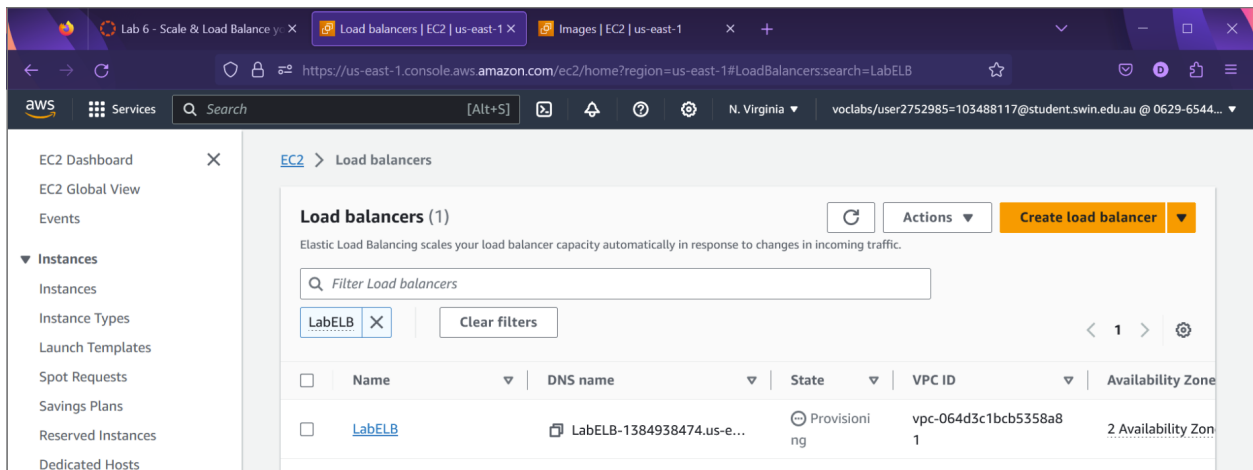


21. Scroll to the bottom and choose Create load balancer

The load balancer is successfully created.

- Choose View load balancer

The load balancer will show a state of *provisioning*. There is no need to wait until it is ready. Please continue with the next task.



Task 3: Create a Launch Template and an Auto Scaling Group

In this task, you will create a *launch template* for your Auto Scaling group. A launch template is a template that an Auto Scaling group uses to launch EC2 instances. When you create a launch template, you specify information for the instances such as the AMI, the instance type, a key pair, and security group.

22. In the left navigation pane, choose **Launch Templates**.

23. Choose Create launch template

24. Configure the launch template settings and create it:

- **Launch template name:** **LabConfig**
- Under **Auto Scaling guidance**, select *Provide guidance to help me set up a template that I can use with EC2 Auto Scaling*
- In the Application and OS Images (Amazon Machine Image) area, choose *My AMIs*.
- **Amazon Machine Image (AMI):** choose *Web Server AMI*
- **Instance type:** choose *t2.micro*
- **Key pair name:** choose *vockey*
- **Firewall (security groups):** choose *Select existing security group*
- **Security groups:** choose *Web Security Group*
- Scroll down to the **Advanced details** area and expand it.
- Scroll down to the **Detailed CloudWatch monitoring** setting. Select *Enable*
Note: This will allow Auto Scaling to react quickly to changing utilization.
- Choose Create launch template

Next, you will create an Auto Scaling group that uses this launch template.

Lab 6 - Scale & Load Balance y

Create launch template | EC2 |

Images | EC2 | us-east-1

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTemplate:

aws Services Search [Alt+S]

N. Virginia

voclabs/user2752985=103488117@student.swin.edu.au @ 0629-6544...

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - required

LabConfig

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '*', '@', ' '.

Template version description

A prod webserver for MyApp

Max 255 chars

Auto Scaling guidance [Info](#)

Select this if you intend to use this template with EC2 Auto Scaling

☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

► Template tags

► Source template

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

▼ Application and OS Images (Amazon Machine Image) - required [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Q Search our full catalog including 1000s of application and OS images

Recents

My AMIs

Quick Start

☒ Owned by me

☐ Shared with me

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

WebServerAMI

ami-011fb8fb8428164c9

2023-10-21T10:32:17.000Z

Virtualization: hvm

ENA enabled: true

Root device type: ebs

boot mode: uefi-preferred

▼ Summary

Software Image (AMI)

Lab AMI for Web Server

ami-011fb8fb8428164c9

Virtual server type (instance type)

t2.micro

Firewall (security group)

Web Security Group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 30 GiB of EBS storage, 2 million IOs, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel

Create launch template

CloudShell

Feedback

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Lab 6 - Scale & Load Balance y

Create launch template | EC2 | x

Images | EC2 | us-east-1

x

+

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTemplate:

aws

Services

Search

[Alt+S]

N. Virginia

voclabs/user2752985=103488117@student.swin.edu.au @ 0629-6544...

▼ Instance type Info

Advanced

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour

On-Demand RHEL base pricing: 0.0716 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

vockey

Create new key pair

▼ Network settings Info

Subnet Info

Don't include in launch template

Create new subnet

When you specify a subnet, a network interface is automatically added to your template.

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Select existing security group

Create security group

Security groups Info

Select security groups

Web Security Group sg-012a22a3a6243daa4

VPC: vpc-064d3c1bcb5358a81

Compare security group rules

► Advanced network configuration

▼ Summary

Software Image (AMI)

Lab AMI for Web Server

ami-011fb8fb8428164c9

Virtual server type (instance type)

t2.micro

Firewall (security group)

Web Security Group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 30 GiB of EBS storage, 2 million IOs, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel

Create launch template

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Advanced details [Info](#)

Purchasing option [Info](#)

☐ Request Spot Instances

If Spot is selected you will not be able to create an Auto Scaling group that spans across multiple pricing options and instance types

IAM instance profile [Info](#)

Don't include in launch template [Create new IAM profile](#)

Hostname type [Info](#)

Don't include in launch template

DNS Hostname [Info](#)

☐ Enable resource-based IPv4 (A record) DNS requests

☐ Enable resource-based IPv6 (AAAA record) DNS requests

Instance auto-recovery [Info](#)

Don't include in launch template

Shutdown behavior [Info](#)

Don't include in launch template

Not applicable for EC2 Auto Scaling

Stop - Hibernate behavior [Info](#)

Don't include in launch template

Not applicable for Amazon EC2 Auto Scaling.

Termination protection [Info](#)

Don't include in launch template

Stop protection [Info](#)

Don't include in launch template

Detailed CloudWatch monitoring [Info](#)

Enable

[Additional charges apply](#)

Summary

Software Image (AMI)

Lab AMI for Web Server
ami-011fb8fb8428164c9

Virtual server type (instance type)

t2.micro

Firewall (security group)

Web Security Group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel **Create launch template**

25. In the Success dialog, choose the **LabConfig** launch template.

EC2 Dashboard

EC2 Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

EC2 > [Launch templates](#) > LabConfig

LabConfig (lt-0716362063d758d71) [Actions](#) [Delete template](#)

Launch template details

Launch template ID	Launch template name	Default version	Owner
lt-0716362063d758d71	LabConfig	1	arn:aws:sts::062965445883:assumed-role/voclabs/user2752985=103488117@student.swin.edu.au

26. From the Actions menu, choose *Create Auto Scaling group*

27. Configure the details in Step 1 (Choose launch template or configuration):

- **Auto Scaling group name:** Lab Auto Scaling Group
- **Launch template:** confirm that the *LabConfig* template you just created is selected.
- Choose Next

The screenshot shows the AWS Management Console interface for creating an Auto Scaling group. The browser tabs include 'Lab 6 - Scale & Load Balance', 'Create Auto Scaling group | EC2', and 'Images | EC2 | us-east-1'. The URL is 'https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateAutoScalingGroup:launchTemplate:'. The page title is 'EC2 > Auto Scaling groups > Create Auto Scaling group'. The left sidebar shows a progress bar with steps 1 through 7. Step 1, 'Choose launch template or configuration', is the active step. The main content area has a heading 'Choose launch template or configuration' with an 'Info' link. Below the heading is a description: 'Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group. If you currently use launch configurations, you might consider migrating to launch templates.' The 'Name' section has a text input field for 'Auto Scaling group name' with the value 'Lab Auto Scaling Group' and a note: 'Must be unique to this account in the current Region and no more than 255 characters.' The 'Launch template' section has a dropdown menu with 'LabConfig' selected and a 'Switch to launch configuration' link. Below this is a 'Create a launch template' link. The 'Version' section has a dropdown menu with 'Default (1)' selected and a 'Create a launch template version' link. The 'Description' section shows a table with details: 'Launch template' (LabConfig, lt-0716362063d758d71), 'Instance type' (t2.micro), 'AMI ID' (ami-011fb8fb8428164c9), 'Security groups' (-), 'Request Spot Instances' (No), 'Key pair name' (vockey), and 'Security group IDs' (sg-012a22a3a6243daa4). The 'Additional details' section shows a table with 'Storage (volumes)' (-) and 'Date created' (Sat Oct 21 2023 17:48:06 GMT+0700 (Indochina Time)). At the bottom right are 'Cancel' and 'Next' buttons.

Step 1
Choose launch template or configuration

Step 2
Choose instance launch options

Step 3 - optional
Configure advanced options

Step 4 - optional
Configure group size and scaling policies

Step 5 - optional
Add notifications

Step 6 - optional
Add tags

Step 7
Review

Choose launch template or configuration [Info](#)

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group. If you currently use launch configurations, you might consider migrating to launch templates.

Name

Auto Scaling group name
Enter a name to identify the group.

Lab Auto Scaling Group

Must be unique to this account in the current Region and no more than 255 characters.

Launch template [Info](#) [Switch to launch configuration](#)

Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

LabConfig

[Create a launch template](#)

Version
Default (1)

[Create a launch template version](#)

Description	Launch template LabConfig lt-0716362063d758d71	Instance type t2.micro
AMI ID ami-011fb8fb8428164c9	Security groups -	Request Spot Instances No
Key pair name vockey	Security group IDs sg-012a22a3a6243daa4	

Additional details

Storage (volumes) -	Date created Sat Oct 21 2023 17:48:06 GMT+0700 (Indochina Time)
------------------------	--

[Cancel](#) [Next](#)

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28. Configure the details in Step 2 (Choose instance launch options):

- **VPC:** choose *Lab VPC*
- **Availability Zones and subnets:** Choose *Private Subnet 1* and then choose *Private Subnet 2*.
- Choose Next

The screenshot shows the AWS Management Console interface for creating an Auto Scaling group. The page is titled "Choose instance launch options" and includes a sidebar with navigation steps. The main content area is divided into sections for instance requirements, network configuration, and availability zones/subnets.

Step 2: Choose instance launch options

Instance type requirements

You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.

[Override launch template](#)

Launch template	Version	Description
LabConfig lt-0716362063d758d71	Default	-

Instance type
t2.micro

Network

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

[vpc-064d3c1bcb5358a81 \(Lab VPC\)](#)
10.0.0.0/16

[Create a VPC](#)

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

[Select Availability Zones and subnets](#)

us-east-1a subnet-0008b7fec878f84b6 (Private Subnet 1) 10.0.1.0/24	X
us-east-1b subnet-09e7c70552953bbf9 (Private Subnet 2) 10.0.3.0/24	X

[Create a subnet](#)

Buttons: Cancel, Skip to review, Previous, Next

29. Configure the details in Step 3 (Configure advanced options):

- Choose **Attach to an existing load balancer**
 - **Existing load balancer target groups**: select *LabGroup*.
- In the **Additional settings** pane:
 - Select **Enable group metrics collection within CloudWatch**
- This will capture metrics at 1-minute intervals, which allows Auto Scaling to react quickly to changing usage patterns.
- Choose Next

The screenshot shows the AWS Management Console interface for the 'Create Auto Scaling group' wizard, specifically Step 3: Configure advanced options. The left sidebar lists the steps: Step 1 (Choose launch template or configuration), Step 2 (Choose instance launch options), Step 3 (optional, Configure advanced options), Step 4 (optional, Configure group size and scaling policies), Step 5 (optional, Add notifications), Step 6 (optional, Add tags), and Step 7 (Review). The main content area is titled 'Configure advanced options - optional' and includes a description: 'Integrate your Auto Scaling group with other services to distribute network traffic across multiple servers using a load balancer or to establish service-to-service communications using VPC Lattice. You can also set options that give you more control over health check replacements and monitoring.'

The 'Load balancing' section has three options: 'No load balancer', 'Attach to an existing load balancer' (selected), and 'Attach to a new load balancer'. The 'Attach to an existing load balancer' option is selected, and the 'Choose from your load balancer target groups' option is also selected. The 'Existing load balancer target groups' section shows a dropdown menu with 'LabGroup | HTTP' selected. The 'Additional settings' section has two options: 'Enable group metrics collection within CloudWatch' (selected) and 'Enable default instance warmup' (not selected).

The screenshot shows the 'Additional settings' section of the 'Create Auto Scaling group' wizard. The 'Monitoring' section has the option 'Enable group metrics collection within CloudWatch' selected. The 'Default instance warmup' section has the option 'Enable default instance warmup' not selected. At the bottom of the wizard, there are buttons for 'Cancel', 'Skip to review', 'Previous', and 'Next'.

30. Configure the details in Step 4 (Configure group size and scaling policies - optional):

- Under **Group size**, configure:
 - **Desired capacity:** 2
 - **Minimum capacity:** 2
 - **Maximum capacity:** 6

This will allow Auto Scaling to automatically add/remove instances, always keeping between 2 and 6 instances running.
- Under **Scaling policies**, choose *Target tracking scaling policy* and configure:
 - **Scaling policy name:** **LabScalingPolicy**
 - **Metric type:** *Average CPU Utilization*
 - **Target value:** **60**

This tells Auto Scaling to maintain an *average* CPU utilization *across all instances* at 60%. Auto Scaling will automatically add or remove capacity as required to keep the metric at, or close to, the specified target value. It adjusts to fluctuations in the metric due to a fluctuating load pattern.
- Choose Next

The screenshot shows the AWS Management Console interface for creating an Auto Scaling group. The browser tabs include 'Lab 6 - Scale & Load Balance', 'Create Auto Scaling group | EC2', and 'Images | EC2 | us-east-1'. The URL is 'https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateAutoScalingGroup:launchTemplate'. The left sidebar shows the navigation menu with steps 1 through 7. Step 4 is highlighted: 'Configure group size and scaling policies'. The main content area is titled 'Configure group size and scaling policies - optional' and includes a description: 'Set the desired, minimum, and maximum capacity of your Auto Scaling group. You can optionally add a scaling policy to dynamically scale the number of instances in the group.' The 'Group size - optional' section has three input fields: 'Desired capacity' (2), 'Minimum capacity' (2), and 'Maximum capacity' (6). The 'Scaling policies - optional' section has a radio button for 'Target tracking scaling policy' (selected) and a 'None' option. Below this, there is a text input for 'Scaling policy name' (LabScalingPolicy), a dropdown for 'Metric type' (Average CPU utilization), a text input for 'Target value' (60), and a text input for 'Instance warmup' (300 seconds). There is also a checkbox for 'Disable scale in to create only a scale-out policy'.

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1
Choose launch template or configuration

Step 2
Choose instance launch options

Step 3 - optional
Configure advanced options

Step 4 - optional
Configure group size and scaling policies

Step 5 - optional
Add notifications

Step 6 - optional
Add tags

Step 7
Review

Configure group size and scaling policies - optional [Info](#)

Set the desired, minimum, and maximum capacity of your Auto Scaling group. You can optionally add a scaling policy to dynamically scale the number of instances in the group.

Group size - optional [Info](#)

Specify the size of the Auto Scaling group by changing the desired capacity. You can also specify minimum and maximum capacity limits. Your desired capacity must be within the limit range.

Desired capacity
2

Minimum capacity
2

Maximum capacity
6

Scaling policies - optional

Choose whether to use a scaling policy to dynamically resize your Auto Scaling group to meet changes in demand. [Info](#)

☒ Target tracking scaling policy
Choose a desired outcome and leave it to the scaling policy to add and remove capacity as needed to achieve that outcome.

☐ None

Scaling policy name
LabScalingPolicy

Metric type [Info](#)
Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

Average CPU utilization

Target value
60

Instance warmup [Info](#)
300 seconds

☐ Disable scale in to create only a scale-out policy

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31. Configure the details in Step 5 (Add notifications - optional):

Auto Scaling can send a notification when a scaling event takes place. You will use the default settings.

- Choose Next

32. Configure the details in Step 6 (Add tags - optional):

Tags applied to the Auto Scaling group will be automatically propagated to the instances that are launched.

- Choose Add tag and Configure the following:

Name: Trac Duc Anh Luong - ID: 103488117

- **Key:** Name
- **Value:** Lab Instance
- Choose Next

The screenshot shows the 'Add tags - optional' step in the AWS console. On the left, a sidebar lists steps 1 through 7. Step 6, 'Add tags', is highlighted. The main area has a title 'Add tags - optional' and a description. A blue information box explains that tags can be added to instances and EBS volumes, but duplicate keys will be overridden. Below this is a 'Tags (1)' section with a table for adding tags. The table has columns for 'Key', 'Value - optional', and 'Tag new instances'. One tag is added with 'Name' as the key and 'Lab Instance' as the value. The 'Tag new instances' checkbox is checked. There is an 'Add tag' button and a '49 remaining' indicator. At the bottom are 'Cancel', 'Previous', and 'Next' buttons.

Key	Value - optional	Tag new instances
Name	Lab Instance	☒

33. Configure the details in Step 6 (Review):

- Review the details of your Auto Scaling group
- Choose Create Auto Scaling group

Your Auto Scaling group will initially show an instance count of zero, but new instances will be launched to reach the **Desired** count of 2 instances.

The screenshot shows the 'Auto Scaling groups' page in the AWS console. A green notification bar at the top states 'Lab Auto Scaling Group, 1 Scaling policy created successfully. Group metrics collection is enabled.' Below this, the 'Auto Scaling groups (1)' section shows a table with one group: 'Lab Auto Scaling Group'. The table has columns for Name, Launch template/configuration, Instances, Status, Desired capacity, and Min. The 'Lab Auto Scaling Group' row shows 0 instances, 'Updating capacity' status, and a desired capacity of 2.

Name	Launch template/configuration	Instances	Status	Desired capacity	Min
Lab Auto Scaling Group	LabConfig Version Default	0	Updating capacity	2	2

Task 4: Verify that Load Balancing is Working

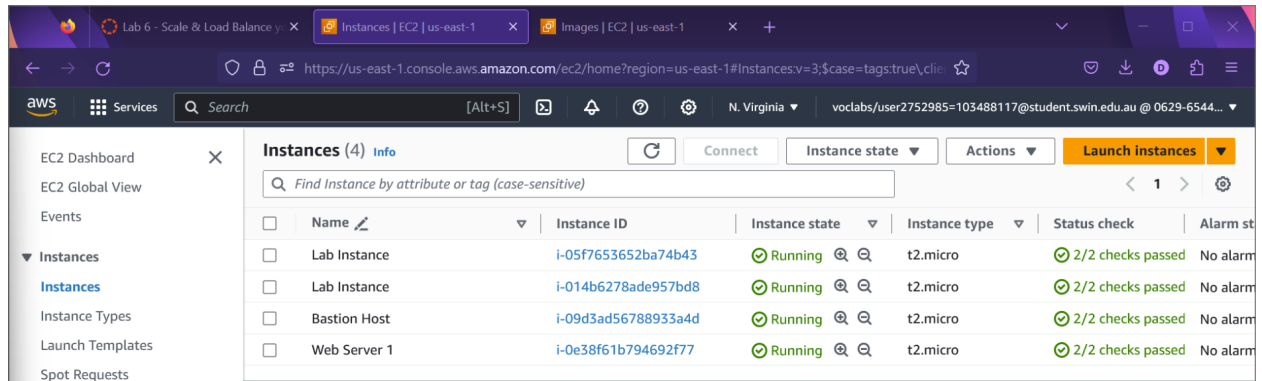
In this task, you will verify that Load Balancing is working correctly.

44. In the left navigation pane, choose **Instances**.

You should see two new instances named **Lab Instance**. These were launched by Auto Scaling.

If the instances or names are not displayed, wait 30 seconds and choose refresh in the top-right.

Next, you will confirm that the new instances have passed their Health Check.



45. In the left navigation pane, choose **Target Groups**.

46. Select *LabGroup*

47. Choose the **Targets** tab.

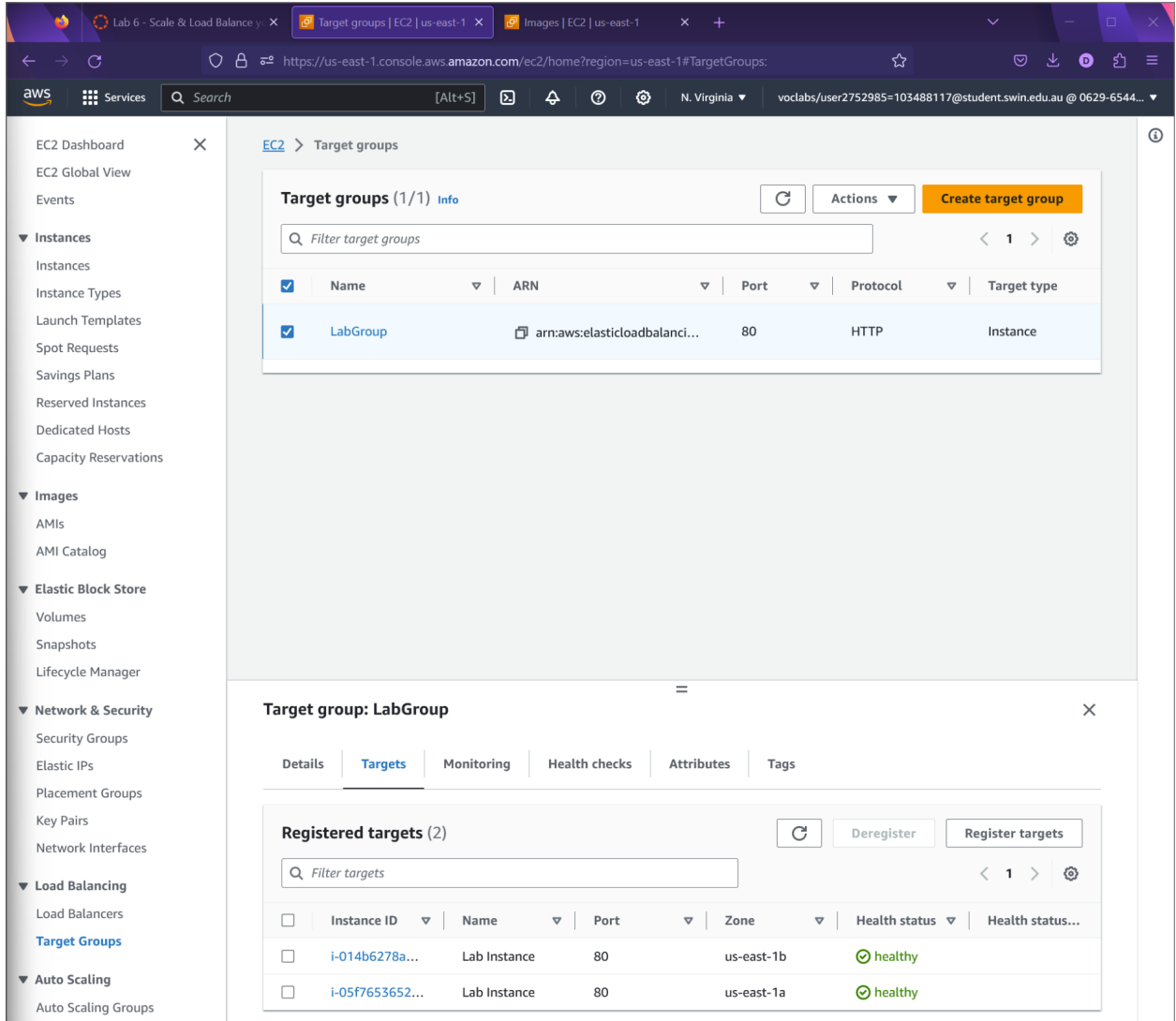
Two target instances named Lab Instance should be listed in the target group.

48. Wait until the **Status** of both instances transitions to *healthy*.

Choose Refresh in the upper-right to check for updates if necessary.

Healthy indicates that an instance has passed the Load Balancer's health check. This means that the Load Balancer will send traffic to the instance.

You can now access the Auto Scaling group via the Load Balancer.



The screenshot displays the AWS Management Console interface. On the left, the navigation pane is expanded to 'Load Balancing' > 'Target Groups'. The main panel shows the 'Target groups (1/1)' list with a table containing one entry, 'LabGroup'. Below this, a modal window titled 'Target group: LabGroup' is open, showing the 'Targets' tab. This tab displays a table of 'Registered targets (2)', both of which are 'healthy'.

Target groups (1/1)										
<table><tr><th>Name</th><th>ARN</th><th>Port</th><th>Protocol</th><th>Target type</th></tr><tr><td>LabGroup</td><td>arn:aws:elasticloadbalanci...</td><td>80</td><td>HTTP</td><td>Instance</td></tr></table>	Name	ARN	Port	Protocol	Target type	LabGroup	arn:aws:elasticloadbalanci...	80	HTTP	Instance
Name	ARN	Port	Protocol	Target type						
LabGroup	arn:aws:elasticloadbalanci...	80	HTTP	Instance						

Target group: LabGroup							
Registered targets (2)							
☐	Instance ID	Name	Port	Zone	Health status	Health status...	
☐	i-014b6278a...	Lab Instance	80	us-east-1b	healthy		
☐	i-05f7653652...	Lab Instance	80	us-east-1a	healthy		

49. In the left navigation pane, choose **Load Balancers**.

50. Select the *LabELB* load balancer.

51. In the Details pane, copy the **DNS name** of the load balancer, making sure to omit "(A Record)".

It should look similar to: *LabELB-1998580470.us-west-2.elb.amazonaws.com*

EC2 Dashboard

EC2 Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

▼ Images

AMIs

AMI Catalog

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

▼ Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

▼ Load Balancing

Load Balancers

Target Groups

▼ Auto Scaling

Auto Scaling Groups

EC2 > Load balancers

Load balancers (1/1)

Actions

Create load balancer

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter Load balancers

☒	Name	DNS name	State	VPC ID	Availability Zones
☒	LabELB	LabELB-1384938474.us-e...	Active	vpc-064d3c1bcb5358a81	2 Availability Zones

Load balancer: LabELB

Details

Listeners and rules

Network mapping

Security

Monitoring

Integrations

Attributes

Tags

Details

Load balancer type	Status	VPC	IP address type
Application	Active	vpc-064d3c1bcb5358a81	IPv4
Scheme	Hosted zone	Availability Zones	Date created
Internet-facing	Z35SXDOTRQ7X7K	subnet-0287c7386b1ea16d2 us-east-1a (use1-az4)	October 21, 2023, 17:40 (UTC+07:00)
		subnet-0ae5dee2c2df397d0 us-east-1b (use1-az6)	
Load balancer ARN		LabELB-1384938474.us-east-1.elb.amazonaws.com (A Record)	
arn:aws:elasticloadbalancing:us-east-1:062965445883:loadbalancer/app/LabELB/7fadabf8339f6b1c			

DNS name copied

CloudShell

Feedback

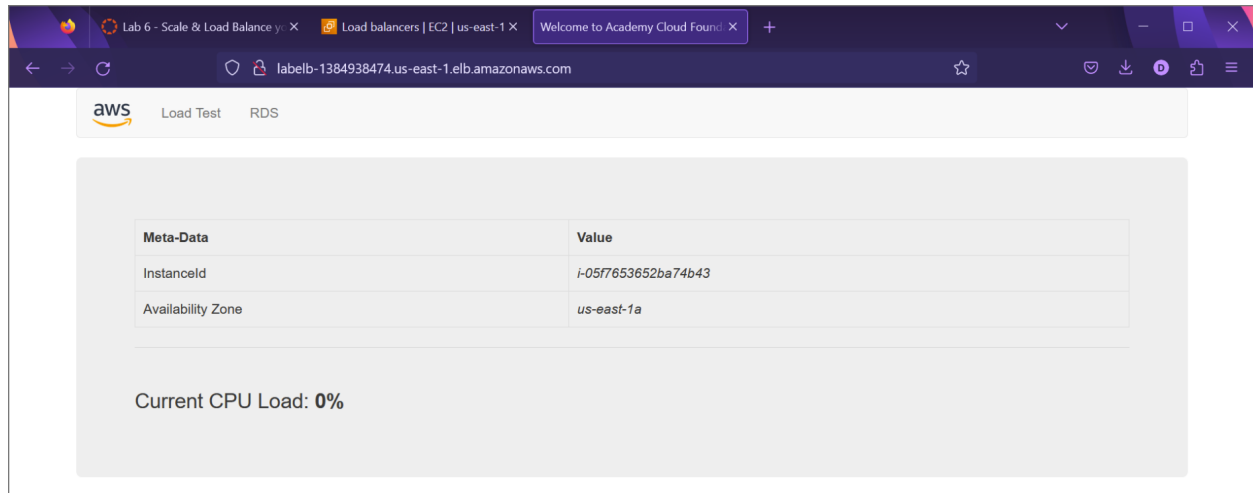
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Privacy

Terms

Cookie preferences

52. Open a new web browser tab, paste the DNS Name you just copied, and press Enter. The application should appear in your browser. This indicates that the Load Balancer received the request, sent it to one of the EC2 instances, then passed back the result.



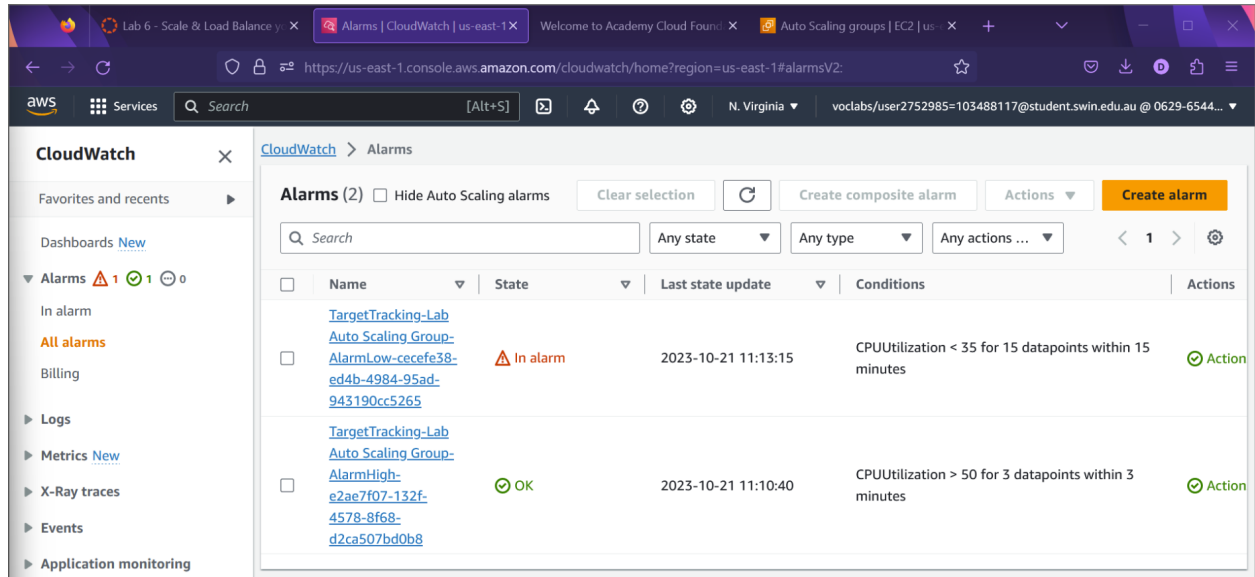
Task 5: Test Auto Scaling

You created an Auto Scaling group with a minimum of two instances and a maximum of six instances. Currently two instances are running because the minimum size is two and the group is currently not under any load. You will now increase the load to cause Auto Scaling to add additional instances.

52. Return to the AWS Management Console, but do not close the application tab — you will return to it soon.
53. in the search box next to Services , search for and select **CloudWatch**.
54. In the left navigation pane, choose **All alarms**.
- Two alarms will be displayed. These were created automatically by the Auto Scaling group. They will automatically keep the average CPU load close to 60% while also staying within the limitation of having two to six instances.
- Note:** Please follow these steps only if you do not see the alarms in 60 seconds.
- On the Services menu, choose **EC2**.
 - In the left navigation pane, choose **Auto Scaling Groups**.
 - Select **Lab Auto Scaling Group**.
 - In the bottom half of the page, choose the **Automatic Scaling** tab.
 - Select **LabScalingPolicy**.
 - Choose Actions and **Edit**.
 - Change the **Target Value** to **50**.
 - Choose Update
 - On the Services menu, choose **CloudWatch**.
 - In the left navigation pane, choose **All alarms** and verify you see two alarms.

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→ I did not see the alarms in 60 seconds. Therefore, I followed the instructions in the Note above and changed the Target Value to 50.



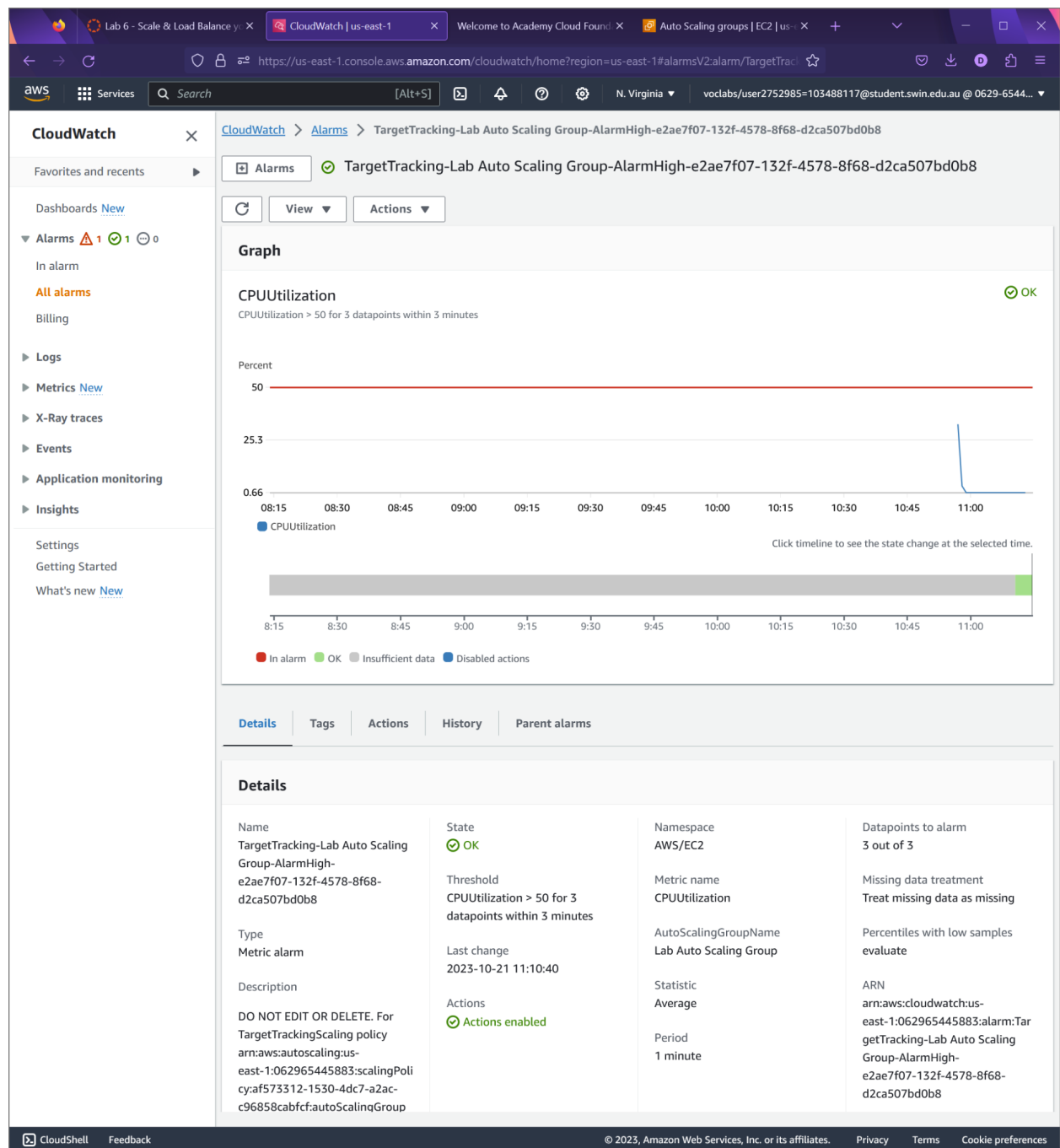
55. Choose the **OK** alarm, which has *AlarmHigh* in its name.

If no alarm is showing **OK**, wait a minute then choose refresh in the top-right until the alarm status changes.

The **OK** indicates that the alarm has *not* been triggered. It is the alarm for **CPU Utilization > 60**, which will add instances when average CPU is high. The chart should show very low levels of CPU at the moment.

You will now tell the application to perform calculations that should raise the CPU level.

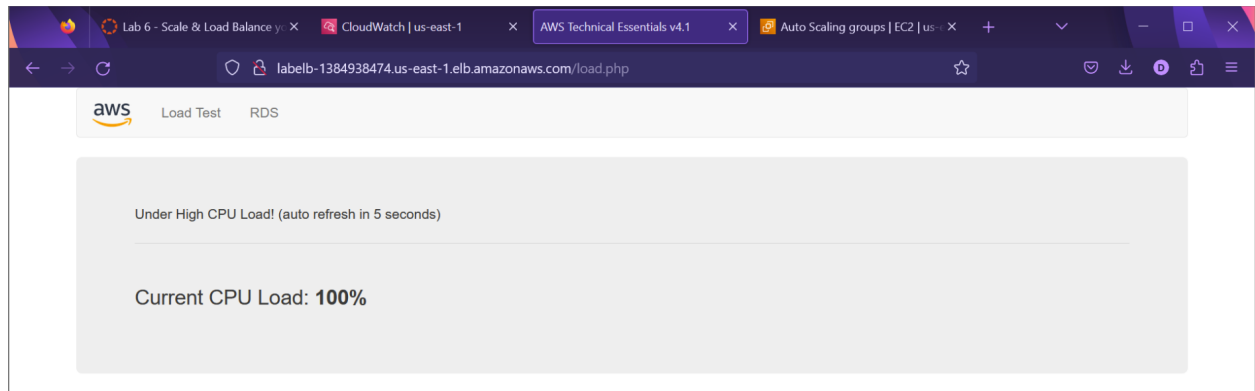
Name: Trac Duc Anh Luong - ID: 103488117



56. Return to the browser tab with the web application.

57. Choose **Load Test** beside the AWS logo.

This will cause the application to generate high loads. The browser page will automatically refresh so that all instances in the Auto Scaling group will generate load. Do not close this tab.

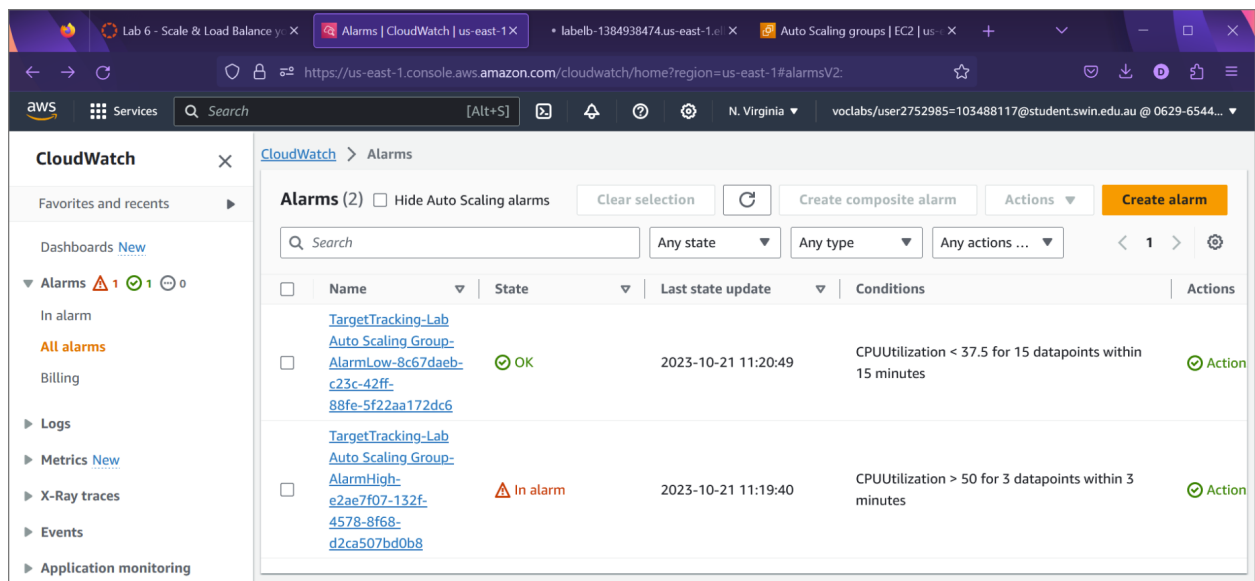


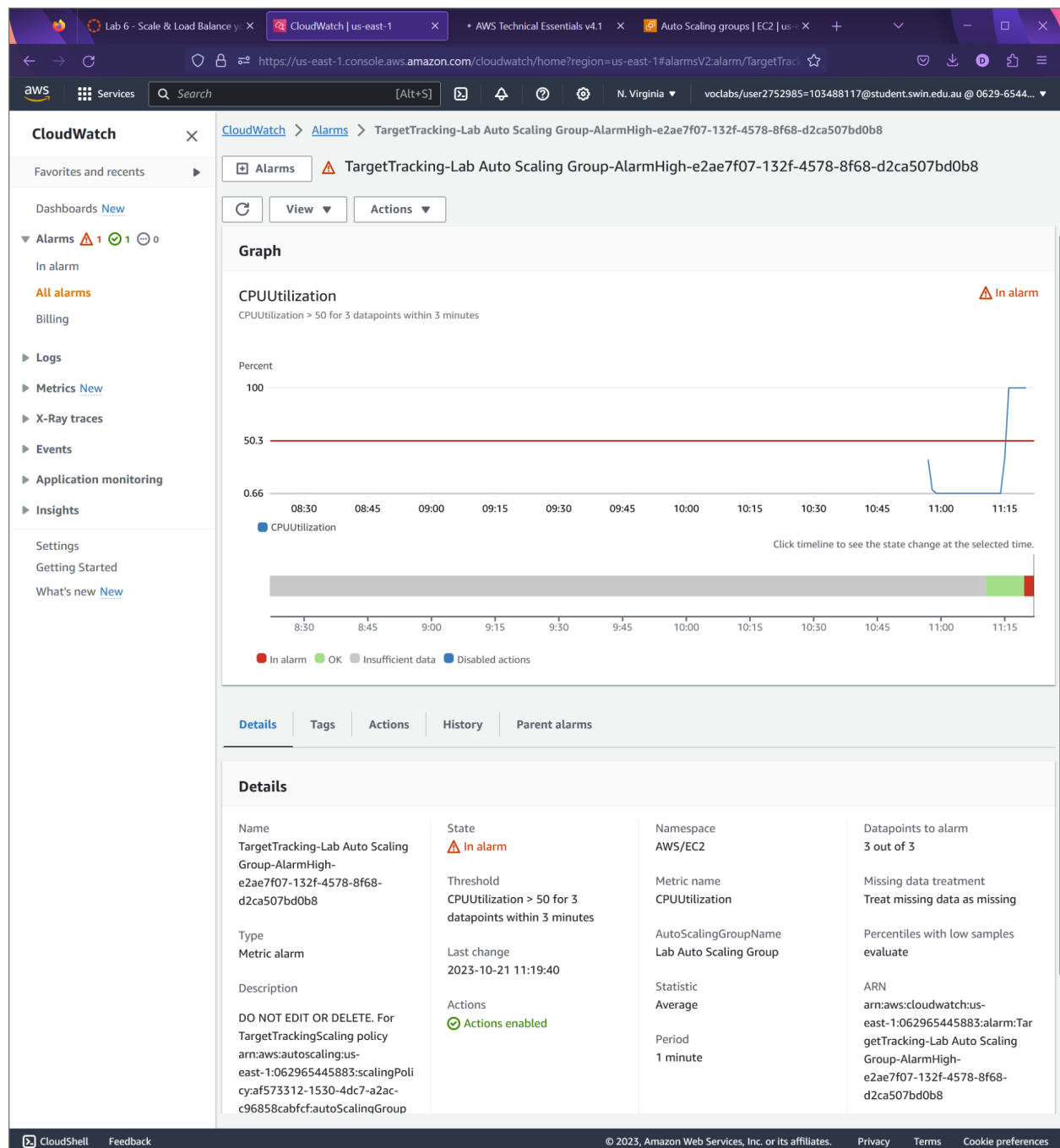
58. Return to browser tab with the **CloudWatch** console.

In less than 5 minutes, the **AlarmLow** alarm should change to **OK** and the **AlarmHigh** alarm status should change to *In alarm*.

You can choose Refresh in the top-right every 60 seconds to update the display.

You should see the **AlarmHigh** chart indicating an increasing CPU percentage. Once it crosses the 60% line for more than 3 minutes, it will trigger Auto Scaling to add additional instances.





59. Wait until the **AlarmHigh** alarm enters the *In alarm* state.

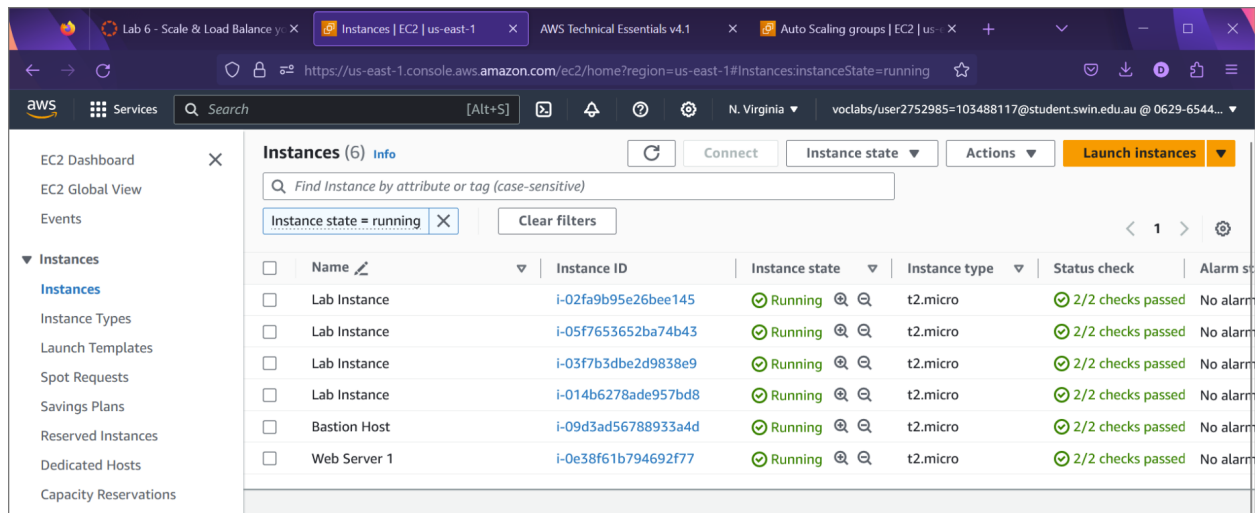
You can now view the additional instance(s) that were launched.

60. In the search box next to Services , search for and select **EC2**.

61. In the left navigation pane, choose **Instances**.

More than two instances labeled **Lab Instance** should now be running. The new instance(s) were created by Auto Scaling in response to the CloudWatch alarm.

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Task 6: Terminate Web Server 1

In this task, you will terminate *Web Server 1*. This instance was used to create the AMI used by your Auto Scaling group, but it is no longer needed.

62. Select **Web Server 1** (and ensure it is the only instance selected).
63. In the Instance state menu, choose **Instance State** > **Terminate Instance**.
64. Choose **Terminate**

